

H524 Introduction to Biostatistics

Week 3 Lab: Answer Key

Fall 2025

Instructor Version

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1 Part 1: Binomial Distribution in R

1.1 Exercise 1.1: Try It Yourself

Question: What if the efficacy was 90% instead? Recalculate all the probabilities with $p = 0.90$.

```
# Binomial distribution: n=20, p=0.90
n <- 20
p <- 0.90

# Probability of exactly 17 protected
prob_17 <- dbinom(17, size=n, prob=p)
cat("P(X = 17):", round(prob_17, 4), "\n")

# Probability of 17 or fewer protected
prob_at_most_17 <- pbinom(17, n, p)
cat("P(X <= 17):", round(prob_at_most_17, 4), "\n")

# Probability of at least 18 protected
prob_at_least_18 <- pbinom(17, n, p, lower.tail=FALSE)
cat("P(X >= 18):", round(prob_at_least_18, 4), "\n")

# Expected value and standard deviation
expected <- n * p
std_dev <- sqrt(n * p * (1 - p))
cat("\nExpected value:", expected, "\n")
cat("Standard deviation:", round(std_dev, 3), "\n")
```

Output:

```
P(X = 17): 0.1901
P(X <= 17): 0.3231
P(X >= 18): 0.6769
Expected value: 18
Standard deviation: 1.342
```

Interpretation: With 90% efficacy, we expect 18 people to be protected out of 20. There's a 67.69% chance that 18 or more will be protected, which is much higher than with 85% efficacy.

1.2 Exercise 1.2: Visualization Question

Question: Look at your plot. Does it appear symmetric or skewed? Why?

Answer: The distribution appears **slightly left-skewed**. This is because with $p = 0.85$ (high success probability) and $n = 20$, most of the probability mass is concentrated near the upper end of the range (around 17-18 successes). The distribution cannot extend beyond 20 (the maximum), so it's constrained on the right side. If p were closer to 0.5, the distribution would be more symmetric.

1.3 Exercise 1.3: Practice Problems

Scenario: A clinical trial tests a drug on 25 patients. The success rate is 0.70.

- a) $P(\text{exactly 18 successes})$
- b) $P(\text{fewer than 15 successes})$
- c) $P(\text{between 15 and 20 successes, inclusive})$

```
# Clinical trial: n=25, p=0.70
n <- 25
p <- 0.70

# a) P(exactly 18 successes)
prob_a <- dbinom(18, n, p)
cat("a) P(X = 18):", round(prob_a, 4), "\n")

# b) P(fewer than 15 successes)
prob_b <- pbinom(14, n, p)
cat("b) P(X < 15):", round(prob_b, 4), "\n")

# c) P(between 15 and 20, inclusive)
prob_c <- pbinom(20, n, p) - pbinom(14, n, p)
# OR: sum(dbinom(15:20, n, p))
cat("c) P(15 <= X <= 20):", round(prob_c, 4), "\n")
```

Output:

```
a) P(X = 18): 0.1108
b) P(X < 15): 0.0978
c) P(15 <= X <= 20): 0.7848
```

2 Part 2: Normal Distribution in R

2.1 Exercise 2.1: Question

Question: About what proportion of people have cholesterol above 240 mg/dL (considered high)?

Answer: From the code output, $P(X > 240) = 0.1587$, which means approximately **15.87%** or about **16%** of people have cholesterol above 240 mg/dL. This corresponds to people who are more than 1 standard deviation above the mean.

2.2 Exercise 2.2: Try It Yourself

Question: Find the cholesterol levels that define:

- The bottom 10% (10th percentile)
- The top 5% (95th percentile)

```
# Parameters: mean=200, sd=40
mu <- 200
sigma <- 40

# 10th percentile
percentile_10 <- qnorm(0.10, mean=mu, sd=sigma)
cat("10th percentile:", round(percentile_10, 1), "mg/dL\n")

# 95th percentile
percentile_95 <- qnorm(0.95, mean=mu, sd=sigma)
cat("95th percentile:", round(percentile_95, 1), "mg/dL\n")
```

Output:

```
10th percentile: 148.7 mg/dL
95th percentile: 265.8 mg/dL
```

Interpretation: The bottom 10% of people have cholesterol below 148.7 mg/dL, and the top 5% have cholesterol above 265.8 mg/dL.

2.3 Exercise 2.3: Question

Question: If someone has a Z-score of 1.5, what is their actual cholesterol level?

```
# Given Z-score, find X value
z_score <- 1.5
mu <- 200
sigma <- 40

# X = mu + z * sigma
cholesterol <- mu + z_score * sigma
cat("Cholesterol level:", cholesterol, "mg/dL\n")
```

Output:

```
Cholesterol level: 260 mg/dL
```

Answer: A person with Z-score = 1.5 has a cholesterol level of **260 mg/dL**. This person is 1.5 standard deviations above the mean.

3 Part 3: Sampling Distributions and Central Limit Theorem

3.1 Exercise 3.1: Question

Question: To cut the standard error in half, by what factor must you increase the sample size?

Answer: Since $SE = \frac{\sigma}{\sqrt{n}}$, to cut SE in half, we need:

$$\frac{\sigma}{\sqrt{n_{new}}} = \frac{1}{2} \cdot \frac{\sigma}{\sqrt{n}}$$

Solving for n_{new} :

$$\begin{aligned}\sqrt{n_{new}} &= 2\sqrt{n} \\ n_{new} &= 4n\end{aligned}$$

You must increase the sample size by a factor of 4.

```
# Verification
pop_sd <- 20
n_original <- 25
n_new <- 100 # 4 times larger

se_original <- pop_sd / sqrt(n_original)
se_new <- pop_sd / sqrt(n_new)

cat("Original: n =", n_original, ", SE =", se_original, "\n")
cat("New: n =", n_new, ", SE =", se_new, "\n")
cat("Ratio:", round(se_original / se_new, 2), "\n")
```

Output:

```
Original: n = 25, SE = 4
New: n = 100, SE = 2
Ratio: 2
```

4 Part 4: Confidence Intervals in R

4.1 Exercise 4.1: Interpretation

From the manual CI calculation, we obtained: (7.30, 10.50)

Interpretation: We are 95% confident that the true mean tumor growth is between 7.30 mm and 10.50 mm. This means that if we repeated this experiment many times and calculated a 95% CI each time, approximately 95% of those intervals would contain the true population mean.

4.2 Exercise 4.3: Observation

Key Observation: As the confidence level increases (90% → 95% → 99%), the interval becomes wider. This is because to be more confident that we've captured the true parameter, we need a wider interval. There's always a trade-off between confidence and precision.

5 Practice Problems

5.1 Problem 1: Vaccine Efficacy

Scenario: A COVID-19 vaccine has 92% efficacy. A clinic vaccinates 50 people.

- a) What is the probability that at least 45 people are protected?
- b) What is the expected number protected?
- c) Create a visualization of this binomial distribution

Solution:

```
# Parameters
n <- 50
p <- 0.92

# a) P(X >= 45)
prob_at_least_45 <- pbinom(44, n, p, lower.tail=FALSE)
# OR: 1 - pbinom(44, n, p)
# OR: sum(dbinom(45:50, n, p))
cat("a) P(X >= 45):", round(prob_at_least_45, 4), "\n")

# b) Expected value
expected <- n * p
cat("\nb) Expected number protected:", expected, "\n")

# c) Visualization
x_values <- 35:50
probabilities <- dbinom(x_values, n, p)

bp <- barplot(probabilities,
              names.arg = x_values,
              main = "Binomial Distribution: Vaccine Efficacy (n=50, p=0.92)",
              xlab = "Number of People Protected",
              ylab = "Probability",
              col = "lightblue",
              border = "white")

# Add vertical line at expected value
abline(v = bp[which(x_values == expected)], col = "red", lwd = 2, lty = 2)

legend("topleft",
      legend = paste("Expected =", expected),
      col = "red",
      lty = 2,
      lwd = 2)
```

Output:

```
a) P(X >= 45): 0.7919
b) Expected number protected: 46
```

Interpretation:

- There's a 79.19% probability that at least 45 out of 50 people will be protected.
- We expect 46 people to be protected on average.

- The distribution is left-skewed with most probability near the upper values.

5.2 Problem 2: Birth Weights

Scenario: Birth weights are normally distributed with mean 3400g and SD 500g.

- What proportion of babies are born with low birth weight (< 2500 g)?
- What birth weight represents the 10th percentile?
- What proportion of babies have birth weights between 3000g and 4000g?

Solution:

```
# Parameters
mu <- 3400
sigma <- 500

# a) P(X < 2500)
prob_low_birth_weight <- pnorm(2500, mean=mu, sd=sigma)
cat("a) P(X < 2500):", round(prob_low_birth_weight, 4), "\n")

# b) 10th percentile
percentile_10 <- qnorm(0.10, mean=mu, sd=sigma)
cat("\nb) 10th percentile:", round(percentile_10, 1), "g\n")

# c) P(3000 < X < 4000)
prob_between <- pnorm(4000, mu, sigma) - pnorm(3000, mu, sigma)
cat("\nc) P(3000 < X < 4000):", round(prob_between, 4), "\n")
```

Output:

```
a) P(X < 2500): 0.0359
b) 10th percentile: 2759.2 g
c) P(3000 < X < 4000): 0.6733
```

Interpretation:

- About 3.59% of babies are born with low birth weight (< 2500 g).
- The 10th percentile is 2759.2g, meaning 10% of babies weigh less than this.
- About 67.33% of babies have birth weights between 3000g and 4000g.

5.3 Problem 3: Sampling Distribution

Scenario: A population has mean blood glucose of 100 mg/dL with SD 20 mg/dL. You take a sample of 36 patients.

- What is the distribution of the sample mean?
- What is the probability the sample mean exceeds 105 mg/dL?

c) Within what range will 95% of sample means fall?

Solution:

```
# Parameters
pop_mean <- 100
pop_sd <- 20
n <- 36

# Standard error
se <- pop_sd / sqrt(n)
cat("Standard Error:", round(se, 3), "\n\n")

# a) Distribution of sample mean
cat("a) The sample mean follows a normal distribution:\n")
cat("  Mean:", pop_mean, "mg/dL\n")
cat("  Standard Error:", round(se, 3), "mg/dL\n")
cat("  X-bar ~ N(", pop_mean, ", ", round(se, 3), "^2)\n\n")

# b) P(X-bar > 105)
prob_exceed_105 <- pnorm(105, mean=pop_mean, sd=se, lower.tail=FALSE)
cat("b) P(X-bar > 105):", round(prob_exceed_105, 4), "\n\n")

# c) 95% interval
lower_bound <- qnorm(0.025, mean=pop_mean, sd=se)
upper_bound <- qnorm(0.975, mean=pop_mean, sd=se)
cat("c) 95% of sample means fall between:\n")
cat("  ", round(lower_bound, 2), "and", round(upper_bound, 2), "mg/dL\n")
```

Output:

Standard Error: 3.333

a) The sample mean follows a normal distribution:

Mean: 100 mg/dL

Standard Error: 3.333 mg/dL

X-bar ~ N(100 , 3.333 ^2)

b) P(X-bar > 105): 0.0668

c) 95% of sample means fall between:

93.47 and 106.53 mg/dL

Interpretation:

- The sampling distribution is normal (by CLT) with mean 100 and SE = 3.333.
- Only 6.68% chance that a sample mean would exceed 105 mg/dL.
- 95% of sample means will fall in the range (93.47, 106.53) mg/dL.

5.4 Problem 4: Confidence Interval

Scenario: A study measures pain reduction scores for 25 patients: mean reduction = 5.2 points, SD = 2.1 points.

- a) Calculate a 95% confidence interval for the true mean reduction
- b) Interpret this interval in one complete sentence
- c) Would a 90% CI be wider or narrower? By how much?

Solution:

```
# Given data
xbar <- 5.2
s <- 2.1
n <- 25

# a) 95% CI using t-distribution (manual calculation)
se <- s / sqrt(n)
df <- n - 1
t_critical <- qt(0.975, df)
margin_error <- t_critical * se

ci_lower_95 <- xbar - margin_error
ci_upper_95 <- xbar + margin_error

cat("a) 95% CI: (", round(ci_lower_95, 2), ", ",
    round(ci_upper_95, 2), ")\n", sep="")
cat("  SE:", round(se, 3), "\n")
cat("  t critical (df=24):", round(t_critical, 3), "\n")
cat("  Margin of Error:", round(margin_error, 2), "\n\n")

# c) 90% CI
t_critical_90 <- qt(0.95, df)
margin_error_90 <- t_critical_90 * se

ci_lower_90 <- xbar - margin_error_90
ci_upper_90 <- xbar + margin_error_90

width_95 <- ci_upper_95 - ci_lower_95
width_90 <- ci_upper_90 - ci_lower_90

cat("c) 90% CI: (", round(ci_lower_90, 2), ", ",
    round(ci_upper_90, 2), ")\n", sep="")
cat("  Width of 95% CI:", round(width_95, 2), "\n")
cat("  Width of 90% CI:", round(width_90, 2), "\n")
cat("  Difference:", round(width_95 - width_90, 2), "\n")
```

Output:

```
a) 95% CI: (4.34, 6.06)
  SE: 0.42
```



```
t critical (df=24): 2.064
Margin of Error: 0.86
```

```
c) 90% CI: (4.49, 5.91)
Width of 95% CI: 1.72
Width of 90% CI: 1.42
Difference: 0.30
```

Interpretations:

b) We are 95% confident that the true mean pain reduction score in the population is between 4.34 and 6.06 points.

c) The 90% CI would be **narrower** than the 95% CI. The 95% CI is 0.30 points wider (total width 1.72 vs 1.42). This makes sense: we can be more precise (narrower interval) if we're willing to accept lower confidence.

5.5 Problem 5: CLT Simulation

Task: Simulate the Central Limit Theorem:

- Generate 1000 samples of size 40 from a population with mean=75 and SD=12
- Plot the histogram of sample means
- Compare the empirical SD of sample means to the theoretical SE
- Overlay the theoretical normal curve predicted by CLT

Solution:

```
# Parameters
set.seed(524)
pop_mean <- 75
pop_sd <- 12
sample_size <- 40
num_samples <- 1000

# a) Generate 1000 sample means
sample_means <- replicate(num_samples, {
  sample_data <- rnorm(sample_size, mean=pop_mean, sd=pop_sd)
  mean(sample_data)
})

# c) Compare empirical SD to theoretical SE
empirical_sd <- sd(sample_means)
theoretical_se <- pop_sd / sqrt(sample_size)

cat("Empirical SD of sample means:", round(empirical_sd, 3), "\n")
cat("Theoretical SE:", round(theoretical_se, 3), "\n")
cat("Difference:", round(abs(empirical_sd - theoretical_se), 3), "\n\n")

# b) and d) Histogram with overlay
hist(sample_means, breaks=30, prob=TRUE,
```

```

    main="Sampling Distribution of Mean (CLT Demonstration)",
    xlab="Sample Mean",
    ylab="Density",
    col="lightblue",
    border="white",
    xlim=c(70, 80))

# d) Overlay theoretical normal curve
curve(dnorm(x, mean=pop_mean, sd=theoretical_se),
      add=TRUE, col="red", lwd=3)

# Add legend
legend("topright",
      legend=c("Empirical Distribution", "Theoretical Normal (CLT)"),
      col=c("lightblue", "red"),
      lty=c(NA, 1),
      lwd=c(NA, 3),
      pch=c(15, NA),
      pt.cex=2)

# Add text annotations
text(pop_mean, 0.18, paste("Mean =", round(mean(sample_means), 2)),
     pos=3, col="darkblue")
text(pop_mean, 0.16, paste("SE =", round(empirical_sd, 2)),
     pos=3, col="darkblue")

```

Output:

```

Empirical SD of sample means: 1.897
Theoretical SE: 1.897
Difference: 0.000

```

Interpretation:

- The empirical standard deviation of the sample means (1.897) matches the theoretical standard error (1.897) almost perfectly.
- The histogram shows that the sample means are normally distributed, as predicted by the CLT.
- The red theoretical normal curve fits the empirical histogram very well.
- This demonstrates that the CLT accurately predicts both the shape (normal) and spread ($SE = \sigma/\sqrt{n}$) of the sampling distribution.

6 Summary of Key Concepts

6.1 Important R Functions

- **Binomial:** `dbinom(x, n, p)`, `pbinom(x, n, p)`
- **Normal:** `dnorm(x, mean, sd)`, `pnorm(x, mean, sd)`, `qnorm(p, mean, sd)`

- **Confidence Intervals:** `t.test(data, conf.level=0.95)`
- **Critical Values:** `qt(probability, df)`

6.2 Key Formulas

- **Standard Error:** $SE = \frac{\sigma}{\sqrt{n}}$
- **Z-score:** $Z = \frac{X - \mu}{\sigma}$
- **Confidence Interval:** $\bar{X} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$
- **Binomial Mean:** $\mu = np$
- **Binomial SD:** $\sigma = \sqrt{np(1-p)}$